RT-Lab Project Overview

UVM Seed Team 5

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Contents

1 Document Description

This document is an overview of four of the RT-Lab projects developed by the UVM SEED team for the Hardware-in-the-Loop for Grid Resilience project. Within this document, a description of each project is provided, including a list of necessary files for the project, the software and hardware requirements for running its simulation, and its inputs and outputs to RT-Lab. After the project description, a detailed explanation of each of the files in the project is provided. The intent of this document is to allow the simulations performed by the team to be replicated. To replicate a specific project download the corresponding folder and follow the OPAL-RT manual. When importing an existing model select the folder and select all files.

2 PQ Base (Offline Simulation)

2.1 Project Description

This project is designed to run an offline simulation of a grid-connected inverter. It is **not** RT-LAB compatible and is intended for testing and debugging the model when RT-LAB is not connected.

2.1.1 Files

The files necessary for this project are:

- ATL_GM_1.m
- PQ_Control_inverter_base.slx

2.1.2 Requirements

The software and hardware required for this project are:

- Matlab 2024b
- Simulink
- Simscape
- Simscape Electrical

2.1.3 Inputs and Outputs for RT-Lab

This project runs offline simulation and contains no OPAL-RT I/O.

2.2 ATL_GM_1.m

This is a MATLAB file containing constants that are used to initialize the SIMULINK model. Ensure that this file is in the same directory as PQ_Control_inverter_base.slx.

2.3 PQ_Control_inverter_base.slx

The `PQ_Control_inverter_base.slx` file contains the Simulink model. The model is divided into two subsystems: the model subsystem `SM_thing` and the control subsystem `SC_thing`. This format mirrors the structure used in the RT-LAB model. The model subsystem contains the inverter, grid, and controller models, while the control subsystem contains the inputs and outputs (power setpoint, fault condition) as well as scopes to observe relevant signals.

The `SM` subsystem consists of four main blocks, described below:

1. **Control Logic**
   - Based on the control schematic presented in this paper.
   - Implements constant PQ control. It is designed to track power setpoints for both real (P) and reactive (Q) power.
   - "Measurements" of voltage and current are taken from the inverter. These signals are processed through a Phase Locked Loop (PLL) and an `abc/dq` transform to obtain direct and quadrature voltage and currents (V_d, V_q, I_d, I_q).
   - These transformed signals are then used in the main control loops and for power calculations.
   - The control schematic consists of two double-nested loops:
     - The first loop contains the active power/current controller.
     - The second loop contains the reactive power/current controller.
   - All controllers are designed as PI controllers with saturation limits.
   - The outputs of the control loops are transformed back to three-phase signals and scaled to create a reference AC signal.
   - This reference signal is used in a Sinusoidal Pulse Width Modulation (SPWM) scheme to generate the switching signals for the inverter.

2. **Inverter**
   - The inverter model is based on the UNIFI inverter model
   - Filter constants for the inverter are defined in the `ATL.m` script.
   - The model uses a Simscape Electrical VSI 5kVA 2 Level Inverter and an LCL filter.

3. **Fault Logic**
   - A Fault Logic block was added to replicate the functionality of ePHASORSIM in offline simulation.
   - The top-level constant represents the per-unit voltage of the grid.
   - The fault logic is controlled from the `SC` subsystem during the simulation runtime.

4. **Grid**
   - The grid is modeled using three unit sine functions, each offset by 120 degrees.
   - These functions are scaled by the per-unit voltage multiplied by the base voltage of the bus.
   - The resulting signals are passed through a Simscape controlled voltage source, and then through a transformer to step the voltage down to 120V.

3 SEED_PSSE_INVERTER_TEST

3.1 Project Description

This project runs software in the loop and can be replicated by downloading the files in the SIL folder from the team Github. It includes ePHASORSIM functionality to model both the inverter and the grid in real time.

3.2 Files

The files necessary for this project are:

- ATL_GM.m
- PQ_control_attempt.slx
- PQ_control_attempt_mdl.mdl
- savnw_conv_32_loads.raw
- savnw.dyr
- phasor03_PSSE_SEED.xlsx

3.2.1 Requirements

- Matlab 2024b
- Simulink
- Simscape
- Simscape Electrical
- RT-Lab
- OP4610IO
- OP5033XG

3.2.2 Inputs and Outputs to RT-Lab

Inputs:
This project contains no Inputs.

Outputs:
The phase 'a' voltage of the capacitor (Analog).

3.3 ATL_GM.m

This is a MATLAB file containing constants that are used to initialize the SIMULINK model. Ensure that this file is in the same directory as PQ_control_attempt.slx.

3.4 PQ_control_attempt.slx

The `SM` subsystem consists of four main blocks, described below:

1. Control Logic

- Based on the control schematic presented in this paper.
- Implements constant PQ control. It is designed to track power setpoints for both real (P) and reactive (Q) power.
- "Measurements" of voltage and current are taken from the inverter. These signals are processed through a Phase Locked Loop (PLL) and an `abc/dq` transform to obtain direct and quadrature voltage and currents (V_d, V_q, I_d, I_q).
- These transformed signals are then used in the main control loops and for power calculations.
- The control schematic consists of two double-nested loops:
  - The first loop contains the active power/current controller.
  - The second loop contains the reactive power/current controller.

- Both controllers are designed as PI controllers with saturation limits.
- The outputs of the control loops are transformed back to three-phase signals and scaled to create a reference AC signal.
- This reference signal is used in a Sinusoidal Pulse Width Modulation (SPWM) scheme to generate the switching signals for the inverter.

2. Inverter

- The inverter model is based on the UNIFI inverter model
- Filter constants for the inverter are defined in the `ATL.m` script.
- The model uses a Simscape Electrical VSI 5kVA 2 Level Inverter and an LCL filter.

3. ePHASORSIM

- The ePHASORSIM block allows for PSS/E integration. The input to the block is a binary condition of a fault. The Output is the per unit bus voltage.

4. Grid

- The grid is modeled using three unit sine functions, each offset by 120 degrees.
- These functions are scaled by the per-unit voltage multiplied by the base voltage of the bus (found in the raw file).
- The resulting signals are passed through a Simscape controlled voltage source, and then through a transformer to step the voltage down to 120V.

3.5 PQ_control_attempt_mdl.mdl

This file is legacy SIMULINK model file that is required for RT-LAB compatibility, but does not provide any additional features to the model.

3.6 savnw_conv_32_loads.raw

This file contains the case data for an example power system network provided by PSS/E, savnw. The savnw case represents a simple transmission system with 23 buses, 6 generators, 4 transformers, and multiple transmission lines and loads. It is commonly used as a baseline for simulation.

To be compatibility with ePHASORSIM, the original savnw case was modified. All loads in the case were originally modeled using constant power (P) and constant current (I) representations, and have been converted to constant impedance (Z) to ensure stability during real-time simulation.

Additionally, the savnw case file has been converted from version 34 to version 32 of PSS/E, which is required by the ePHASORSIM solver. A one-line diagram of the savnw case is shown in Figure 1.

3.7 savnw.dyr

The savnw.dyr file contains the dynamic model data associated with the savnw case. This file defines the behavior of dynamic components such as generators, exciters, governors, etc., which are each represented by a dynamic model with parameters that describe its response to system disturbances over time.

3.8 phasor03_PSSE_SEED.xlsx

The phasor03_PSSE_SEED.xlsx file is a spreadsheet of inputs and outputs to the ePHASORSIM solver block in RT-Lab, telling the solver which signals to accept and return. In this case, the input is a three-phase bus-to-ground fault on bus 101, and the output is the voltage at bus 101.

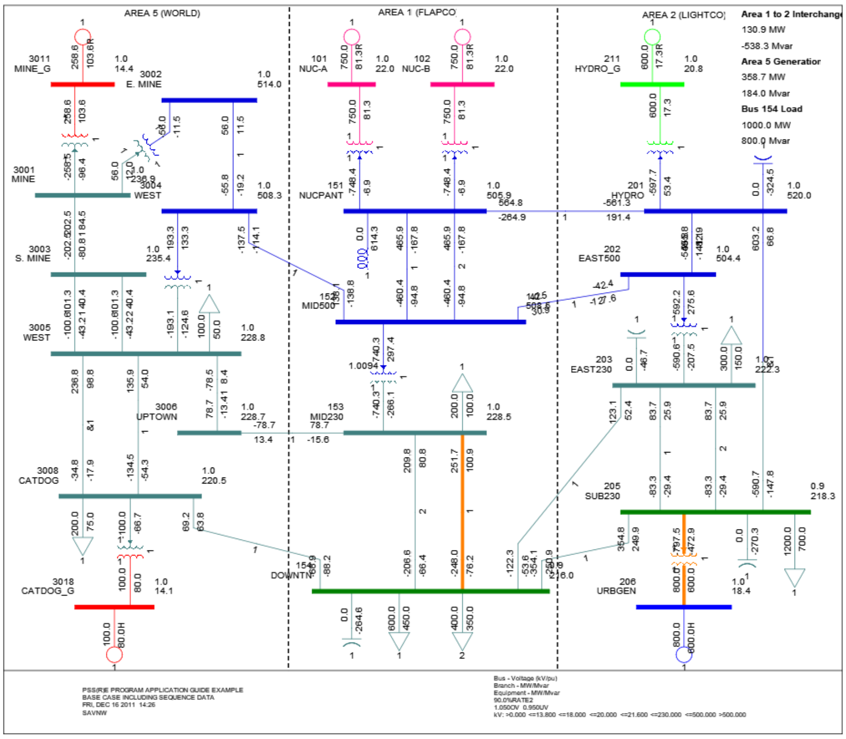

Figure 1: Example PSS/E Case savnw.raw

4 SEED_CHIL_TEST_2

4.1 Project Description

This project runs Control-Hardware-in-the-Loop (CHIL) and can be replicated by downloading the files in the Mega CHIL folder in the team Github. It includes both ePHASORSIM functionality (incorporating a simulated grid model) and code/configuration for an Arduino Mega microcontroller to act as the inverter controller.

4.1.1 Files

The files necessary for this project are:

- ATL_GM.m
- CHIL_TEST.slx
- CHIL_TEST_mdl.mdl
- savnw_conv_32_loads.raw
- savnw.dyr
- savnw_IO.xlsx
- PQ_Control.ino

4.1.2 Requirements

- Matlab 2024b
- Simulink
- Simscape
- Simscape Electrical

- Arduino IDE 2.1.0
- RT-Lab
- OP4610IO
- OP5033XG
- Arduino MEGA

4.1.3 Inputs and Outputs to RT-Lab

Inputs:
Three digital inverter switching signals and their inverses (digital):

- a
- a'
- b
- b'
- c
- c'

The scaled real and reactive power as calculated from the Arduino (analog):

- P_out_raw
- Q_out_raw

Outputs:
The inverter model signal measurements required for PQ control.

- Vd
- Vq
- Id
- Iq
- Pref
- theta

4.2 ATL_GM.m

This is a MATLAB file containing constants that are used to initialize the SIMULINK model. Ensure that this file is in the same directory as PQ_Control_inverter_base.slx.

4.3 CHIL_TEST.slx

1. Inverter

- The inverter model is based on the UNIFI inverter model
- Filter constants for the inverter are defined in the `ATL.m` script.
- The model uses a Simscape Electrical VSI 5kVA 2 Level Inverter and an LCL filter.

2. ePHASORSIM

- The ePHASORSIM block allows for PSS/E integration.
- The input is a multiplexed binary fault signal for 3 busses from the .xlsx file.
- The output is a multiplexed per unit signal for the 3 busses from the .xlsx file.
- The inverter was connected to bus 101 by demuxing the PU voltages and using the first value.

3. Grid

- The grid is modeled using three unit sine functions, each offset by 120 degrees.
- These functions are scaled by the per-unit voltage multiplied by the base voltage of bus 101 (found in the raw file).
- The resulting signals are passed through a Simscape controlled voltage source, and then through a transformer to step the voltage down to 120V.

4.4 CHIL_TEST_mdl.mdl

This file is legacy SIMULINK model file that is required for RT-LAB compatibility, but does not provide any additional features to the model.

4.5 savnw_conv_32_loads.raw

This file contains the case data for an example power system network provided by PSS/E, savnw. The savnw case represents a simple transmission system with 23 buses, 6 generators, 4 transformers, and multiple transmission lines and loads. It is commonly used as a baseline for simulation.

To be compatibility with ePHASORSIM, the original savnw case was modified. All loads in the case were originally modeled using constant power (P) and constant current (I) representations, and have been converted to constant impedance (Z) to ensure stability during real-time simulation.

Additionally, the savnw case file has been converted from version 34 to version 32 of PSS/E, which is required by the ePHASORSIM solver. A one-line diagram of the savnw case is shown in Figure 1.

4.6 savnw.dyr

The savnw.dyr file contains the dynamic model data associated with the savnw case. This file defines the behavior of dynamic components such as generators, exciters, governors, etc., which are each represented by a dynamic model with parameters that describe its response to system disturbances over time.

4.7 savnw_IO.xlsx

The savnw_IO.xlsx file is a spreadsheet of inputs and outputs to the ePHASORSIM solver block in RT-Lab, telling the solver which signals to accept and return. In this case, the inputs are three, three-phase bus-to-ground faults on busses 101, 102, and 3008, and the outputs are the voltages at busses 101, 102, and 3008.

4.8 PQ_Control.ino

This file is an Arduino code file that can be uploaded to the Arduino Mega to run PQ control for the simulated inverter. This file must be downloaded into a folder with the name PQ_Control.ino and opened in the Arduino IDE, then uploaded to the Arduino Mega, which must be connected to the laptop through a USB port.

Tables 1 – 3 present the inputs and outputs to the Arduino Mega for this project.

Name	Pin	Description
`Vd_pin`	A0	d-axis voltage (V_d)
`Vq_pin`	A1	q-axis voltage (V_q)
`Id_pin`	A2	d-axis current (I_d)
`Iq_pin`	A3	q-axis current (I_q)
`freq_pin`	A4	Grid frequency input
`theta_pin`	A5	Phase angle (θ) input
`P_ref_pin`	A6	Reference active power (P_{ref})

Table 1: Analog Inputs

Name	Pin	Description
`PWM_A`	2	Phase A upper switch control
`PWM_B`	3	Phase B upper switch control
`PWM_C`	4	Phase C upper switch control
`PWM_A_o`	5	Phase A lower switch control (inverted)
`PWM_B_o`	6	Phase B lower switch control (inverted)
`PWM_C_o`	7	Phase C lower switch control (inverted)

Table 2: Digital Outputs (PWM signals)

Name	Pin	Description
`P_OUT`	10	Real power output (P) to OPAL-RT
`Q_OUT`	11	Reactive power output (Q) to OPAL-RT

Table 3: Analog Outputs

5 SEED_CHIL_DUE_TEST

5.1 Project Description

This project runs Control-Hardware-in-the-Loop (CHIL) and can be replicated by downloading the files in the Due CHIL folder in the team Github. It includes both ePHASORSIM functionality (incorporating a simulated grid model) and code/configuration for an Arduino Due microcontroller to act as the inverter controller. The Due is faster and more powerful than the Mega.

5.1.1 Files

The files necessary for this project are:

- ATL_GM.m
- CHIL_TEST.slx
- CHIL_TEST_mdl.mdl
- savnw_conv_32_loads.raw
- savnw.dyr
- savnw_IO.xlsx
- FULL_SEND_DUE_PLEASE_WORK_PLEASE.ino

5.1.2 Requirements

- Matlab 2024b
- Simulink
- Simscape
- Simscape Electrical

- Arduino IDE 2.1.0
- RT-Lab
- OP4610IO
- OP5033XG
- Arduino DUE

5.1.3 Inputs and Outputs to RT-Lab

Inputs:
PWM signals for each phase and the scaled real and reactive power calculated from the Arduino (analog):

- pwm_a
- pwm_b
- pwm_c
- P_out_raw
- Q_out_raw

Outputs:
The inverter model signal measurements required for PQ control.

- Vd
- Vq
- Id
- Iq
- Pref
- theta

5.2 ATL_GM.m

This is a MATLAB file containing constants that are used to initialize the SIMULINK model. Ensure that this file is in the same directory as PQ_Control_inverter_base.slx.

5.3 CHIL_TEST.slx

1. Inverter

- The inverter model is based on the UNIFI inverter model
- Filter constants for the inverter are defined in the `ATL.m` script.
- The model uses a Simscape Electrical VSI 5kVA 2 Level Inverter and an LCL filter.

2. ePHASORSIM

- The ePHASORSIM block allows for PSS/E integration.
- The input is a multiplexed binary fault signal for 3 busses from the .xlsx file.
- The output is a multiplexed per unit signal for the 3 busses from the .xlsx file.
- The inverter was connected to bus 101 by demuxing the PU voltages and using the first value.

3. Grid

- The grid is modeled using three unit sine functions, each offset by 120 degrees.
- These functions are scaled by the per-unit voltage multiplied by the base voltage of bus 101 (found in the raw file).
- The resulting signals are passed through a Simscape controlled voltage source, and then through a transformer to step the voltage down to 120V.

5.4 CHIL_TEST_mdl.mdl

This file is legacy SIMULINK model file that is required for RT-LAB compatibility, but does not provide any additional features to the model.

5.5 savnw_conv_32_loads.raw

This file contains the case data for an example power system network provided by PSS/E, savnw. The savnw case represents a simple transmission system with 23 buses, 6 generators, 4 transformers, and multiple transmission lines and loads. It is commonly used as a baseline for simulation.

To be compatibility with ePHASORSIM, the original savnw case was modified. All loads in the case were originally modeled using constant power (P) and constant current (I) representations, and have been converted to constant impedance (Z) to ensure stability during real-time simulation.

Additionally, the savnw case file has been converted from version 34 to version 32 of PSS/E, which is required by the ePHASORSIM solver. A one-line diagram of the savnw case is shown in Figure 1.

5.6 savnw.dyr

The savnw.dyr file contains the dynamic model data associated with the savnw case. This file defines the behavior of dynamic components such as generators, exciters, governors, etc., which are each represented by a dynamic model with parameters that describe its response to system disturbances over time.

5.7 savnw_IO.xlsx

The savnw_IO.xlsx file is a spreadsheet of inputs and outputs to the ePHASORSIM solver block in RT-Lab, telling the solver which signals to accept and return. In this case, the inputs are three, three-phase bus-to-ground faults on busses 101, 102, and 3008, and the outputs are the voltages at busses 101, 102, and 3008.

5.8 FULL_SEND_DUE_PLEASE_WORK_PLEASE.ino

This file is an Arduino code file that can be uploaded to the Arduino Due to run PQ control for the simulated inverter. This file must be downloaded into a folder with the name FULL_SEND_DUE_PLEASE_WORK_PLEASE.ino and opened in the Arduino IDE, then uploaded to the Arduino Due, which must be connected to the laptop through a USB port.

Name	Pin	Description
Vd_pin	A0	D-axis voltage input
Vq_pin	A1	Q-axis voltage input
Id_pin	A2	D-axis current input
Iq_pin	A3	Q-axis current input
P_ref_pin	A6	Real power reference input
freq_pin	A4	Frequency input
theta_pin	A5	Theta input (used for PLL phase angle)

Table 4: Analog Inputs to PQ Control Script

Name	**Pin**	**Description**
PWM_A	2	PWM Output for Phase A
PWM_B	3	PWM Output for Phase B
PWM_C	4	PWM Output for Phase C
PWM_A_o	5	Inverse PWM Output for Phase A
PWM_B_o	6	Inverse PWM Output for Phase B
PWM_C_o	7	Inverse PWM Output for Phase C
P_OUT	10	Analog output representing real power (P)
Q_OUT	11	Analog output representing reactive power (Q)

Table 5: PWM and Analog Outputs from PQ Control Script